Qcontrol – a software package for solving the quantum control problem

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Working experience summary

Scientific Research

- Computational and analytical **molecular spectroscopy** (academic) – more than 15 years, 20 scientific articles, PhD degree, international team (Europe, Russia, USA)
- **Quantum control problem** (academic) – 2 years, Hebrew University + HIT, solo development

Software Engineering

Commercial (industry)

- Improving quantum chemistry modeling tools – Intel inc. R&D group (2 years)
- Deep Learning software improvement, implementing new algos – AXELTEC (2 years)

Research (academic)

- Long-term support, debugging and improving an old legacy codebase
- Writing a modeling software package from scratch

Formulation of the problem

- **Direct problem:** study of a quantum system dynamics driven by an external electromagnetic field – **well described** by a unitary transformation generated by a time-dependent Hamiltonian
- **Inverse problem:** finding the field that generates a specific unitary transformation – a more **sophisticated** problem
- **What is given:**
  - Quantum system type and parameters
  - Initial state
  - Desired final state
  - Initial guess for the laser field with parameters to be adjusted
- **What to find:** optimal parameters of the laser field required for moving the system from the initial state to the final one

Propagation: Newtonian polynomial algorithm with Chebyshev-based interpolation scheme

❖ **General problem:**

Time-dependent Schrodinger equation (TDSE): $i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi$ ⇒ $\psi = \psi(0)\exp\left(\frac{-i\hat{H}t}{\hbar}\right)$

How to calculate an "exponent from tensor"?

❖ **Method:**

Consider $f(z) = \exp\left(-\frac{izt}{\hbar}\right)$ and a set of interpolation points $\{z_k, f_k\}\,|\,f_k = f(z_k), z_k \in \mathbb{R}$.

The interpolation Newtonian polynomials R_n are used to approximate $f(z)$:

$$f(z) \approx P_N(z) = \sum_{n=0}^{N} a_n R_n(z);\; R_0(z) = 1, R_n(z) = \prod_{j=0}^{n-1}(z - z_j);\;\; a_0 = f_0, a_1 = \frac{f_1 - f_0}{z_1 - z_0}, a_n = \frac{f_n - P_{n-1}(z_n)}{R_n(z_n)}.$$

For a Chebyshev-based interpolation scheme, z_n are chosen as the zeros of the *N*+1-th Chebyshev polynomial (this ensures stability of a_n): $z_k = 2\cos\left(\frac{2k+1}{2N}\pi\right) \in [-2;2]$ ⇒

⇒ we should reflect it in the spectrum of $\hat{H}$ (assuming that it fits into a range $[\epsilon_{min}, \epsilon_{max}]$):

$\hat{H}' = 4\frac{\hat{H} - \epsilon_{min}\hat{I}}{\epsilon_{max} - \epsilon_{min}} - 2\hat{I}$ ⇒ the function $f(z')$ in a_n must be calculated at the points

$$f\left[\frac{1}{4}(\epsilon_{max} - \epsilon_{min})(z+2) + \epsilon_{min}\right] = \exp\left(-izt_d\right)\exp\left(-it_s\right), t_d = \frac{(\epsilon_{max} - \epsilon_{min})t}{4\hbar}, t_s = \frac{(\epsilon_{max} + \epsilon_{min})t}{2\hbar}.$$

Propagation: Newtonian polynomial algorithm with Chebyshev-based interpolation scheme

❖ Recursion scheme (step-by-step algorithm):

- ➢ Estimation of minimum and maximum limits of the Hamiltonian spectrum $[\epsilon_{min}, \epsilon_{max}]$;
- ➢ Calculating N_{Ch} (a power of 2) interpolation points $z_k = 2\cos\theta_k$;
- ➢ Obtaining $\hat{H}' = 4\dfrac{\hat{H} - \epsilon_{min}\hat{I}}{\epsilon_{max} - \epsilon_{min}} - 2\hat{I}$;
- ➢ Calculating the shift operation $f(z) \to f(z') = \exp(-iz't_d)$ and the phase shift in the final wavefunction $\exp(-it_s) = \exp\left(-2i\dfrac{\epsilon_{max} + \epsilon_{min}}{\epsilon_{max} - \epsilon_{min}} \cdot t_d\right)$;
- ➢ Recursion:
$$\Phi_0 = \psi(0)$$
$$\Phi_1 = (\hat{H}' - z_0\hat{I})\Phi_0$$
$$\cdots$$
$$\Phi_{n+1} = (\hat{H}' - z_n\hat{I})\Phi_n$$
- ➢ Calculating divided difference coefficients a_n ;
- ➢ Final wavefunction $\Psi = \exp(-it_s)\sum_{n=0}^{N-1} a_n\Phi_n$;
- ➢ Improving stability of the algorithm by numerically reordering of the interpolation points set;
- ➢ Explicit dependency $\hat{H}(t)$ ⇒ using small time steps so that $\hat{H}$ is almost stationary within each step.

Coordinate grids, basis functions and kinetic energy operator

- **Spatial grid:** symmetric, equidistant, centered at the minimum of the potential, number of the grid points N_p – a power of 2.
- **Time grid:** equidistant, number of the grid points N_t.
- **Single-particle basis functions:**
  - harmonic oscillators $\psi(x) = \frac{1}{\sqrt{a\sqrt{\pi}}} \exp\left(-\frac{(x-x_0)^2}{2a^2} + ip_0 x\right)$;
  - Morse oscillators

$$\psi(x) = \sqrt{a(2\beta-1)}\sqrt{\frac{(2\beta)^{2\beta-1}}{\Gamma(2\beta)}} \exp\left(-\beta \exp(-a(x-x_0))\right) \exp\left(-a(\beta-1/2)(x-x_0)\right),\ \beta = \frac{2D_e}{\omega_e},\ \omega_e = a\sqrt{\frac{2D_e}{m}}$$

$x_0,\ p_0$ – initial coordinate and momentum, a – scaling factor, m – reduced mass of the system,

ω_e – harmonic frequency, D_e – dissociation energy, $x_e = \frac{1}{2\beta}$ – anharmonicity factor

- **Kinetic energy operator** calculations:
  - FFT: $\psi(x_i) \rightarrow \bar{\psi}_k(k_i) = a_i$
  - $\bar{\phi}(k_i) = \hat{T}_k \bar{\psi}_k(k_i) = \frac{\hbar^2 k_i^2}{2m} a_i$
  - IFFT: $\bar{\phi}(k_i) \rightarrow \phi(x_i) = \hat{T}_{ij}\psi(x_i)$

Potentials and laser fields

- Harmonic potential: $V = \frac{k_s(x-x_0)^2}{2}, k_s = \frac{1}{ma^4}$ – a stiffness coefficient;
- Morse potential: $V = D_e(1-\exp(-a(x-x_0)))^2$
- Laser field types:
  - Constant: $E_L = E_L^0$
  - Time-dependent: $E_L = E_L^0 f(t) g(t)$, E_L^0– a constant amplitude, $f(t)$- envelope, $g(t)$– high-frequency part
  - $f(t)$ types:
    - Gaussian: $f(t) = \exp\left(-\frac{(t-t_0)^2}{2\sigma^2}\right)$
    - Square sin: $f(t) = \sin^2\left(\frac{2\pi(t-t_0)}{\sigma}\right)$
    - Maxwell: $f(t) = \frac{t^2}{\sigma^2}\exp\left(-\frac{(t-t_0)^2}{2\sigma^2}\right)$
  - $g(t)$ types:
    - Exponent: $g(t) = \exp(ip\omega_L t)$
    - Cos: $g(t) = \cos(Cp\omega_L t)$
    - Sin: $g(t) = \sin(Cp\omega_L t)$
    - Set of cos: $g(t) = w_0\cos(p\omega_L t) + \sum_{j=2k+1}^{C}(w_j\cos(p\omega_L tj) + w_{j+1}\cos(p\omega_L t/j))$
    - Set of sin: $g(t) = w_0\sin(p\omega_L t) + \sum_{j=1}^{C} w_j\sin(p\omega_L t(2j-1))$

Hamiltonian types

- **Conventional type:**
  - for $\Psi = \begin{pmatrix} \psi_e(x,t) \\ \psi_g(x,t) \end{pmatrix}$: $\hat{H} = \begin{pmatrix} V_e & -E_L^* \\ -E_L & V_g \end{pmatrix} + \frac{\hat{p}^2}{2m}$
  - for $\Psi = \begin{pmatrix} \psi_e^\omega(x,t) \\ \psi_g^\omega(x,t) \end{pmatrix}$, $\psi_{g/e} = \psi_{g/e}^\omega \cdot \exp\left(\pm i \frac{\omega_L}{2} t\right)$: $\hat{H} = \begin{pmatrix} V_e - \frac{\hbar\omega_L}{2} & -E_L^0 \\ -E_L^0 & V_g + \frac{\hbar\omega_L}{2} \end{pmatrix} + \frac{\hat{p}^2}{2m}$
- **Two-level type:**

$$\hat{H} = \begin{pmatrix} \frac{V_0}{2} & -E_L^* \\ -E_L & -\frac{V_0}{2} \end{pmatrix}, V_0 = \hbar\omega_0$$

- **Bose-Hubbard-like in angular momentum form:**
  - With laser field in *z* component:

$$\hat{H} = -2\Delta \hat{J}_x + 2U\hat{J}_z^2 + 2E_L^0(t)\hat{J}_z$$

  - With laser field in *x* component:

$$\hat{H} = -2\Delta \hat{J}_x E_L^0(t) + 2W\hat{J}_z^2 + 2U\hat{J}_z$$

  - For qubit-like system with $U < 0$:

$$\hat{H} = -\Delta(\hat{J}_+ E_L(t) + \hat{J}_- E_L^*(t)) + 2W\hat{J}_z^2 + 2U\hat{J}_z$$

$$\hat{J}_z^2 = \sum_{i=0}^{N_{levs}-1} (\hat{a}_i^+ \hat{a}_i)^2$$

$$\hat{J}_x = \sum_{i=0}^{N_{levs}-1} \frac{\hat{a}_{i+1}^+ \hat{a}_i + \hat{a}_{i-1}^+ \hat{a}_i}{2} = \frac{\hat{J}_+ + \hat{J}_-}{2}$$

Quantum control types

- **Intuitive control:** a sequence of identical laser pulses transfers a quantum system from the initial to the final state, and then back to the initial one.
- **Local control**
  - **population control:** transition from the ground state to the excited one under the influence of external laser field with controlled envelope form $E_L^0 f(t)$, with the target operator $\hat{O} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.
  - **projection control:** transition from the ground state to the excited one under the influence of external laser field with controlled chirp via fitting of the frequency multiplier parameter p, with the target operator $\hat{O} = |\psi_g^\omega >< \psi_g^\omega| + |\psi_e^\omega >< \psi_e^\omega| = \begin{pmatrix} 1+|S_{ge}|^2 & 2S_{ge} \\ 2S_{ge}^* & 1+|S_{ge}|^2 \end{pmatrix}$, $S_{ge} =< \psi_g^\omega|\psi_e^\omega >$.
- **Optimal control** (Krotov algorithm): transition from the ground state to the excited one under the influence of external laser field with controlled envelope form using an iterative algorithm, when the propagation on a current time step is happening partially under the old field from the previous time step, and partially – under the new field, which is calculated "on the fly".
  - **Conventional Hamiltonian form** with the projection target operator $\hat{O} = |\psi_e >< \psi_e|$.
  - **General unitary transformation** with an Hadamard target operator

$$\hat{O} = \frac{1}{\sqrt{N_{levs}}} \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \bar{\omega} & \bar{\omega}^2 & \dots & \bar{\omega}^{N_{levs}-1} \\ 1 & \bar{\omega}^2 & \bar{\omega}^4 & \dots & \bar{\omega}^{2(N_{levs}-1)} \\ \dots & \dots & \dots & \dots & \dots \\ 1 & \bar{\omega}^{N_{levs}-1} & \bar{\omega}^{2(N_{levs}-1)} & \dots & \bar{\omega}^{(N_{levs}-1)(N_{levs}-1)} \end{pmatrix}, \bar{\omega} = \exp\left(\frac{2\pi i}{N_{levs}}\right)$$

Local control

- The algorithms are based on a non-iterative, unidirectional technique that calculates the control field on-the-fly based on the instantaneous state of the system.

- A target operator $\hat{O}$ represents the desired final state. The algorithm focuses on increasing the expectation value of this operator over time.

- The local control method is constructed to ensure that the target operator's expectation value $< \hat{O} >_{\psi^\omega} = < \psi^\omega(t) | \hat{O} | \psi^\omega(t) >$ increases monotonically, driving the system toward the desired target state.

- Heisenberg equation of motion of the operator $\hat{O}$: $\frac{d < \hat{O} >_{\psi^\omega}}{dt} = \frac{i}{\hbar} < [\hat{H}, \hat{O}] >_{\psi^\omega}$. This gives the local control conditions:
  - for population target operator: $\frac{d < \hat{O} >_{\psi^\omega}}{dt} = \frac{2}{\hbar} E_L^0 \boxed{f(t)} \Im(< \psi_g^\omega | \psi_e^\omega >) > 0$
  - for projection target operator: $\frac{d < \hat{O} >_{\psi^\omega}}{dt} = -\frac{4}{\hbar} \Im(S_{ge}(V_g - V_e)_{ge} + \hbar \boxed{p} \omega (S_{ge})^2) > 0, S_{ge} = < \psi_g^\omega | \psi_e^\omega >, (V_g - V_e)_{ge} = < \psi_g^\omega | V_g - V_e | \psi_e^\omega >$

- In the code both the cases were implemented using Euler algorithm with additional decay term to improve convergence features:
  - if $\frac{d < \hat{O} >_{\psi^\omega}}{dt} \equiv \dot{\hat{O}} < 0$, the desired value of the laser field parameter (envelope form or frequency multiplier) on the current step g_{goal} is obtained from the "good" positive $\dot{\hat{O}}$ value from the previous step and current wavefunctions
  - Acceleration term: $\Delta g_{vel} = -\left(c_1 (g - g_{goal}) + c_2 g_{vel} \left(\frac{g_{goal}}{g} \right)^q \right) \Delta t;\ c_1, c_2, q$ – input parameters
  - Euler step: $\Delta g = g_{vel} \Delta t$

Optimal control

- **Krotov's method** is an optimal control algorithm for quantum systems that guarantees monotonic convergence using sequential updates with tailored control fields. It employs iterative forward and backward propagations to locally update control parameters while effectively handling constraints and nonlinear dynamics.
- **Conventional Hamiltonian form** with the projection target operator
  - Problem: $i\hbar\frac{\partial\chi}{\partial t} = \hat{H}\chi, t \in [0,T]$ $\chi(x,T) = \hat{P}_e\psi(x,T)$ $i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi, \psi(x,0) = \psi_0(x)$
  - Goal functional (stopping conditions for the algorithm):
    $$J = \left|\sum_k < \hat{O}\psi_k^i|\psi_k(T) >\right| = \left|\sum_k < \psi_k^f|\psi_k(T) >\right| \equiv |\tau| \overset{\max}{\rightarrow} N_{levs} = 2$$
  - Step-by-step algorithm:
    - Initial guess for the laser field: $E_L^{(0)}(t)$
    - Forward propagation: $\psi_0(x) \overset{E_L^{(0)}}{\rightarrow} \psi(x,t) \overset{E_L^{(0)}}{\rightarrow} \psi(x,T)$
    - Obtaining $\chi(x,T) =< \psi_f(x)|\psi(x,T) > \psi_f(x)$
    - Backward propagation: $\chi(x,0) \overset{\bar{E}_L^{(0)}}{\leftarrow} \chi(x,t) \overset{\bar{E}_L^{(0)}}{\leftarrow} \chi(x,T)$
    - Obtaining laser field on the *k*-th step:
      $$E_L^{k+1}(t) = -\frac{2}{\hbar\lambda}\Im\left[< \chi_e^{\omega(k)}(t)|\psi_g^{\omega(k+1)}(t) >\right] \qquad \bar{E}_L^{k}(t) = -\frac{2}{\hbar\lambda}\Im\left[< \chi_e^{\omega(k)}(t)|\psi_g^{\omega(k)}(t) >\right]$$

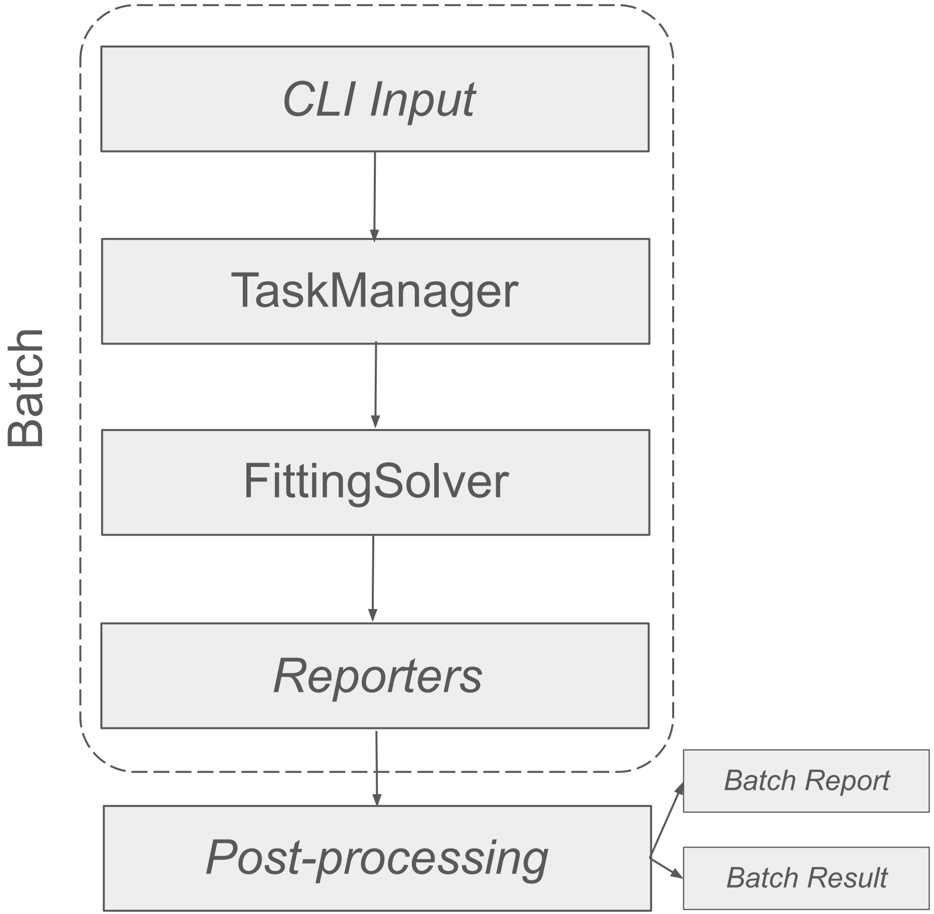
Optimal control

- **General unitary transformation** with an Hadamard target operator
  - Problem (for a 2-level system): $\psi_0^i = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \psi_1^i = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ $\hat{O} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ $t \in [0, T]$ $\psi_k^f = \hat{O}\psi_k^i$
  - Goal functionals (stopping conditions for the algorithm):
    - $F_{sm} = -\left| \sum_k < \psi_k^f | \psi_k(T) > \right|^2 = -|\tau|^2 \overset{\min}{\rightarrow} -N_{levs}^2 = -4$
    - $F_{re} = -\Re \left[\sum_k < \psi_k^f | \psi_k(T) > \right] \overset{\min}{\rightarrow} -N_{levs} = -2$
    - $F_{ss} = -\sum_k \left| < \psi_k^f | \psi_k(T) > \right|^2 \overset{\min}{\rightarrow} -N_{levs} = -2$
  - Step-by-step algorithm:
    - Initial guess for the laser field: $E_L^{(0)}(t) = E_0 f(t)$
    - Backward propagation: $\chi_k^{(n)}(x, 0) \overset{E_L^{(n)}}{\leftarrow} \chi_k^{(n)}(x, t) \overset{E_L^{(n)}}{\leftarrow} \psi_k^f(x)$
    - Obtaining $a_{sm}^{(n)} = \sum_k < \psi_k^i(x) | \chi_k^{(n)}(x, 0) > \sim F_{sm}$ or $a_{ss_k}^{(n)} = < \psi_k^i(x) | \chi_k^{(n)}(x, 0) > \sim F_{ss}$ or $a_{re}^{(n)} = \frac{1}{2} \sim F_{re}$
    - Forward propagation, *l*-th time step: $\psi_k^{(n+1)}((l-1)\Delta t) \overset{E_L^{(n+1)}(l)}{\rightarrow} \psi_k^{(n+1)}(l\Delta t)$

$$\Delta E_L^{(n)}(l\Delta t - \Delta t/2) = -\frac{f(l\Delta t - \Delta t/2)}{\hbar\lambda} \Im \left[\sum_k a_k^{(n)} < \chi_k^{(n)}((l-1)\Delta t) | \psi_k^{(n+1)}((l-1)\Delta t) > \right], l \in [1, N_t], \psi_k^{(n+1)}(0) = \psi_k^i(x)$$

$$E_L^{(n+1)}(l\Delta t - \Delta t/2) = E_L^{(n)}(l\Delta t - \Delta t/2) + \Delta E_L^{(n)}(l\Delta t - \Delta t/2)$$

General code architecture



TaskManager interface implementations:

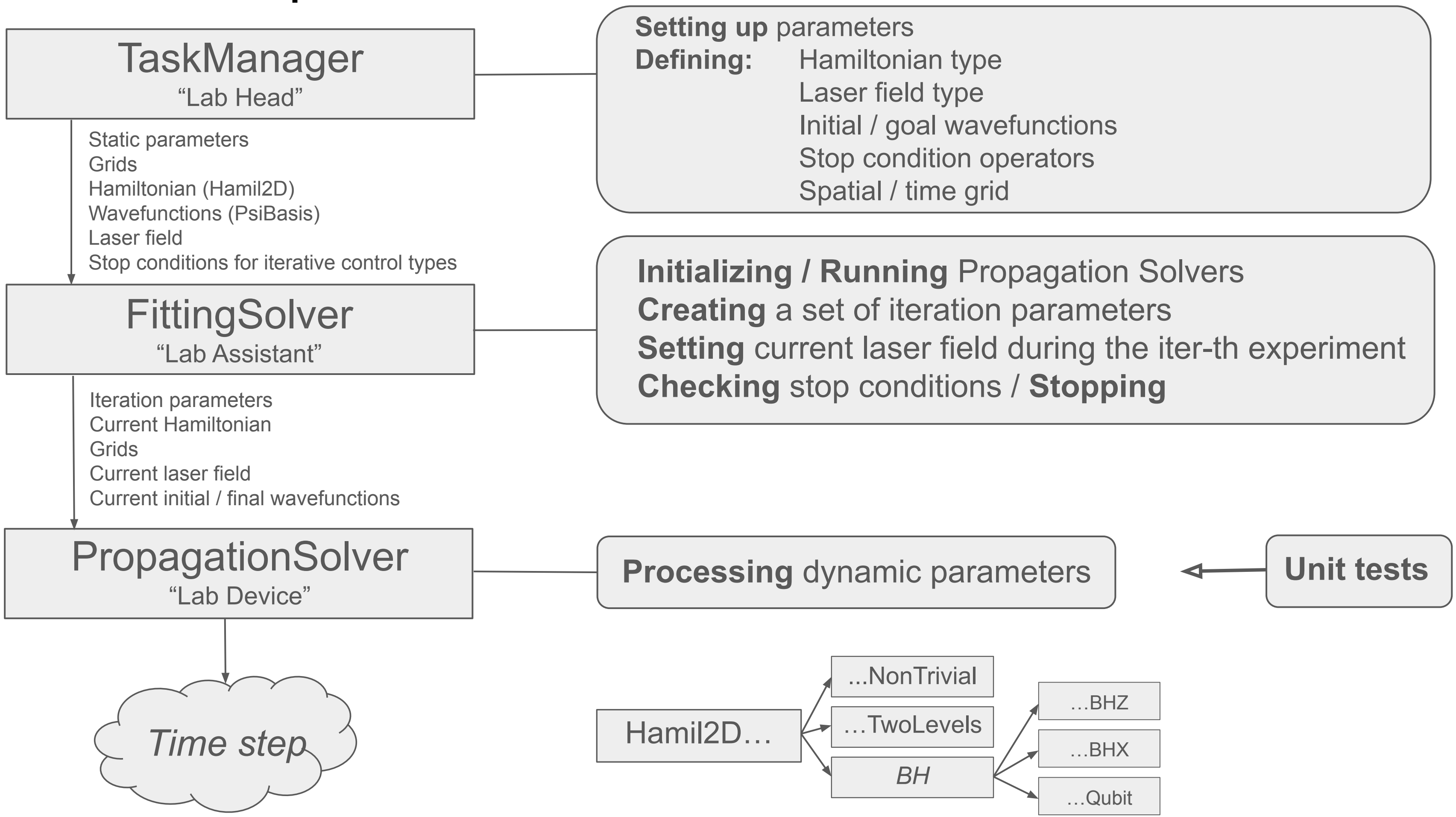
- `HarmonicSingleStateTaskManager`
- `HarmonicMultipleStateTaskManager`
- `MorseSingleStateTaskManager`
- `MorseMultipleStateTaskManager`
- `MultipleStateUnitTransformTaskManager`

Functional tests: 12 in total

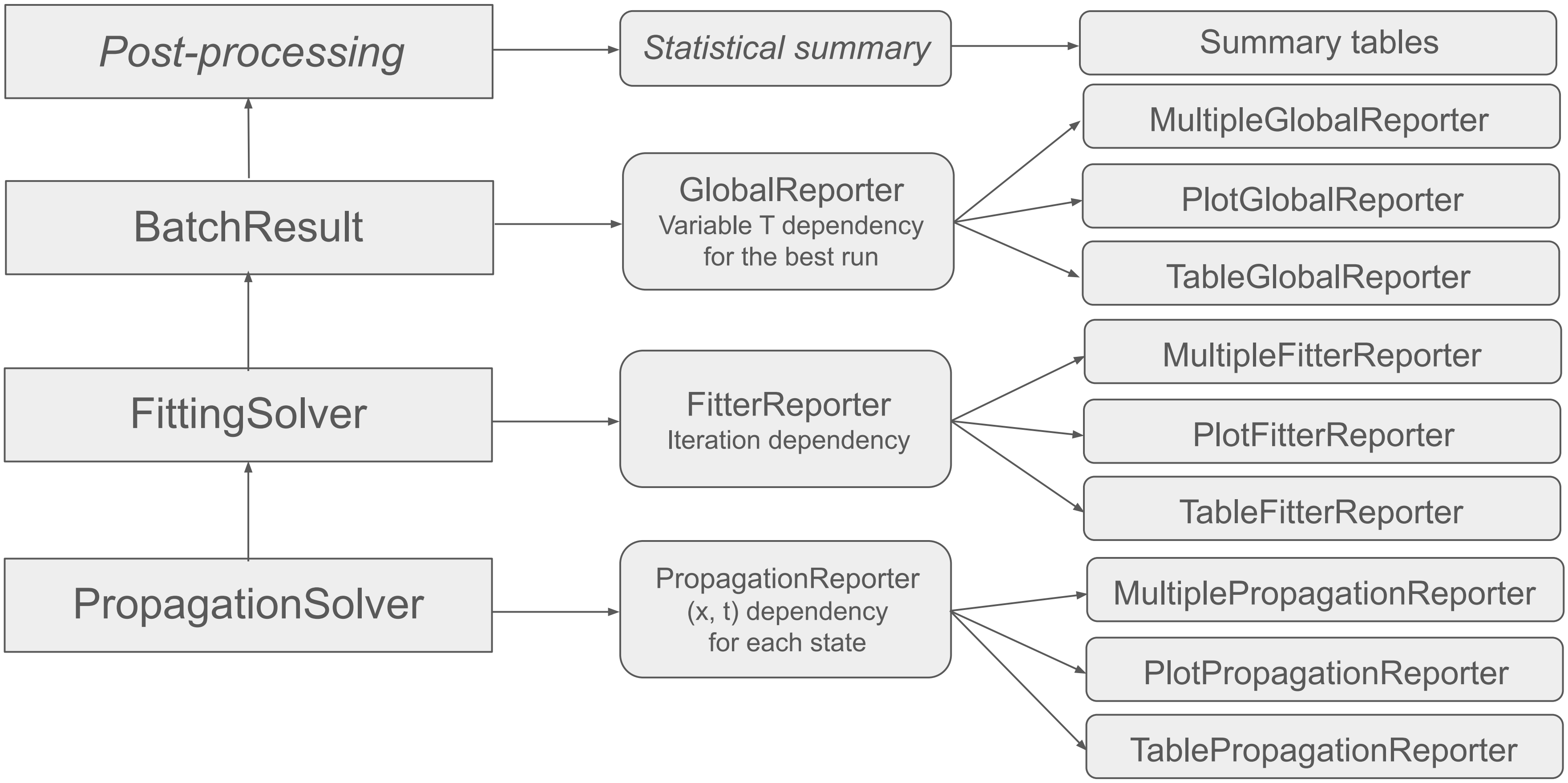
Reporter “flavours”:

- Tables
- Plots (html + js)

Calculation part architecture



Reporting part architecture



Configuration files

input_task.json:

```
{
 "run_id": "ut/run1",
 "task_type": "optimal_control_unit_transform",
 "hamil_type": "BH_model",
 "lf_aug_type": "z",
 "init_guess": "sqrsin",
 "init_guess_hf": "cos_set",
 "nb": 2,
 "T": 0.5E-13,
 "U": 5.0,
 "delta": 25.0,
 "sigma_auto": true,
 "nu_L_auto": true,
 "fitter": {
   "epsilon": 1e-3,
   "impulses_number": 1,
   "iter_max": 2000,
   "h_lambda": 0.005,
   "pcos": 4.0,
   "hf_hide": false,
   "Em": 5.0,
   "propagation": {
     "nch": 8,
     "t0": 0.0,
     "E0": 40.0,
     "nt": 3500
   },
   "mod_log": 500
 }
}
```

$$\sigma = 2T$$

$$\nu_L = \frac{1}{2T}$$

`plotting_flag` parameter selects different reporting options: ALL, TABLES, TABLES_ITER, PLOTS, NONE

input_report.json:

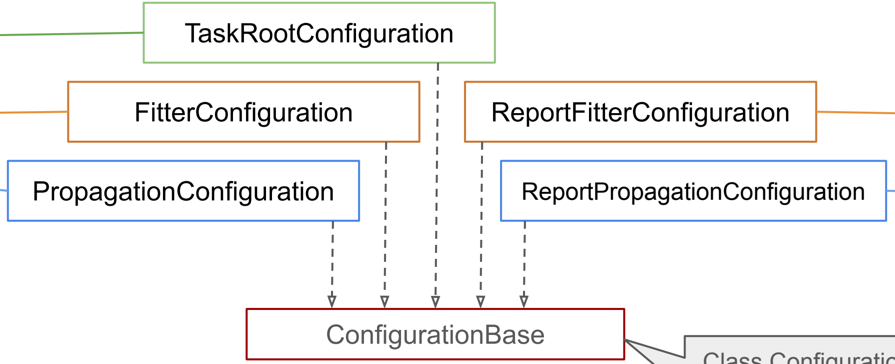
```
{
 "fitter": {
   "table_glob_path": "../output_ut/run1/glob.csv",
   "out_path": "../output_run1/T=0.5E-13",
   "plotting_flag": "tables_iter",

   "table.imin": -1,
   "table.imod_fileout": 1,
   "table.tab_iter": "tab_iter.csv",
   "table.tab_iter_E": "tab_iter_E.csv",

   "plot.imin": -1,
   "plot.imod_plotout": 1,
   "plot.inumber_plotout": 15,
   "plot.gr_iter": "fig_iter_goal.html",
   "plot.gr_iter_E": "fig_gr_iter_E.html",
   "plot.gr_iter_F": "fig_gr_iter_F.html",

   "propagation": {
     "table.lmin": 0,
     "table.mod_fileout": 1,
     "table.tab_abs": "tab_abs_{level}.csv",
     "table.tab_real": "tab_real_{level}.csv",
     "table.tab_tvals": "tab_tvals_{level}.csv",
     "table.tab_tvals_fit": "tab_tvals_fit.csv",

     "plot.lmin": 0,
     "plot.mod_plotout": 10,
     "plot.mod_update": 20,
     "plot.number_plotout": 15
   }
 }
}
```



Class ConfigurationBase contains redefinition of the dot operator to a field, so that each input parameter can be reached as a class member, i.e. `conf_task.fitter.propagation.iter_max`

Configuration templates

input_task.json:

```
{
 "run_id": { "@SUBST:LIST":
 [ "ut/run1", "ut/run2", "ut/run3", "ut/run4",
   "ut/run5", "ut/run6", "ut/run7", "ut/run8",
   "ut/run9", "ut/run10"
 ] },
 "task_type": "optimal_control_unit_transform",
 "hamil_type": "BH_model",
 "lf_aug_type": "z",
 "init_guess": "sqrsin",
 "init_guess_hf": "cos_set",
 "nb": 2,
 "T": { "@SUBST:LIST":
 [
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   1.0E-13, 1.1E-13, 1.2E-13, 1.3E-13, 1.4E-13,
   1.5E-13, 1.6E-13, 1.7E-13, 1.8E-13, 1.9E-13,
   2.0E-13, 2.1E-13, 2.2E-13, 2.3E-13, 2.4E-13,
   2.5E-13, 2.6E-13, 2.7E-13, 2.8E-13, 2.9E-13,
   3.0E-13, 3.1E-13, 3.2E-13, 3.3E-13, 3.4E-13,
   3.5E-13, 3.6E-13, 3.7E-13, 3.8E-13, 3.9E-13
 ] },
 "U": 5.0,
 "delta": 25.0,
 "sigma_auto": true,
 "nu_L_auto": true,
 "fitter": {
   "epsilon": 1e-3,
   "impulses_number": 1,
   "iter_max": 2000,
   "h_lambda": 0.005,
   "pcos": 4.0,
   "hf_hide": false,
   "Em": 5.0,
   "propagation": {
     "nch": 8,
     "t0": 0.0,
     "E0": 40.0,
     "nt": 3500
   },
   "mod_log": 500
 }
}
```
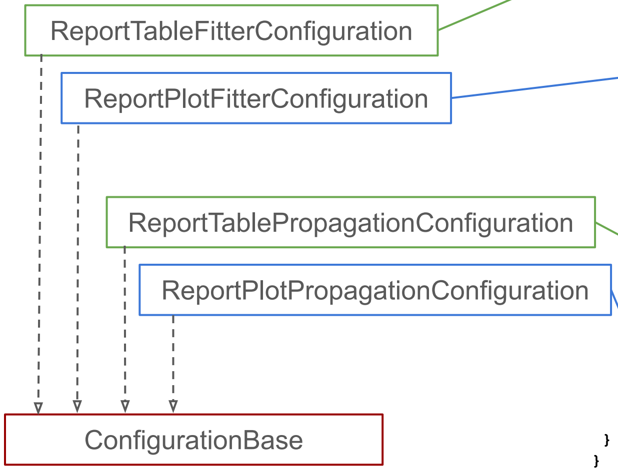

DSL:
These templates are processed before the run. A single item is chosen for each run configuration json



input_report.json:

```
{
 "fitter": {
   "table_glob_path": "../output_{input:$.run_id}/glob.csv",
   "out_path": "../output_{input:$.run_id}/T={input:$.T}",
   "plotting_flag": "tables_iter",

   "table.imin": -1,
   "table.imod_fileout": 1,
   "table.tab_iter": "tab_iter.csv",
   "table.tab_iter_E": "tab_iter_E.csv",

   "plot.imin": -1,
   "plot.imod_plotout": 1,
   "plot.inumber_plotout": 15,
   "plot.gr_iter": "fig_iter_goal.html",
   "plot.gr_iter_E": "fig_gr_iter_E.html",
   "plot.gr_iter_F": "fig_gr_iter_F.html",

   "propagation": {
     "table.lmin": 0,
     "table.mod_fileout": 1,
     "table.tab_abs": "tab_abs_{level}.csv",
     "table.tab_real": "tab_real_{level}.csv",
     "table.tab_tvals": "tab_tvals_{level}.csv",
     "table.tab_tvals_fit": "tab_tvals_fit.csv",

     "plot.lmin": 0,
     "plot.mod_plotout": 10,
     "plot.mod_update": 20,
     "plot.number_plotout": 15
   }
 }
}
```

These templates are processed during the run, according to the chosen CLI parameters

Supported task types

- ❖ **`task type = single pot`**
  Simple propagation of an initial harmonic / Morse wavefunction (without transition)

- ❖ **`task type = filtering`**
  Filtering task – a behaviour of an initial single harmonic oscillator in a Morse potential (without transition)

- ❖ **`task type = trans wo control`**
  Calculation of transition from the ground state to the excited one under the influence of external laser field with gaussian envelope and a constant chirp (without control)

- ❖ **`task type = intuitive control`**
  Calculation of transitions from the ground state to the excited state and back to the ground one under the influence of a sequence of equal laser pulses with gaussian envelopes and constant chirps

- ❖ **`task type = local control population`**
  Calculation of transition from the ground state to the excited one under the influence of external laser field with controlled envelope form by the local control algorithm with population target operator

- ❖ **`task type = local control projection`**
  Calculation of transition from the ground state to the excited one under the influence of external laser field with controlled chirp by the local control algorithm with projection target operator

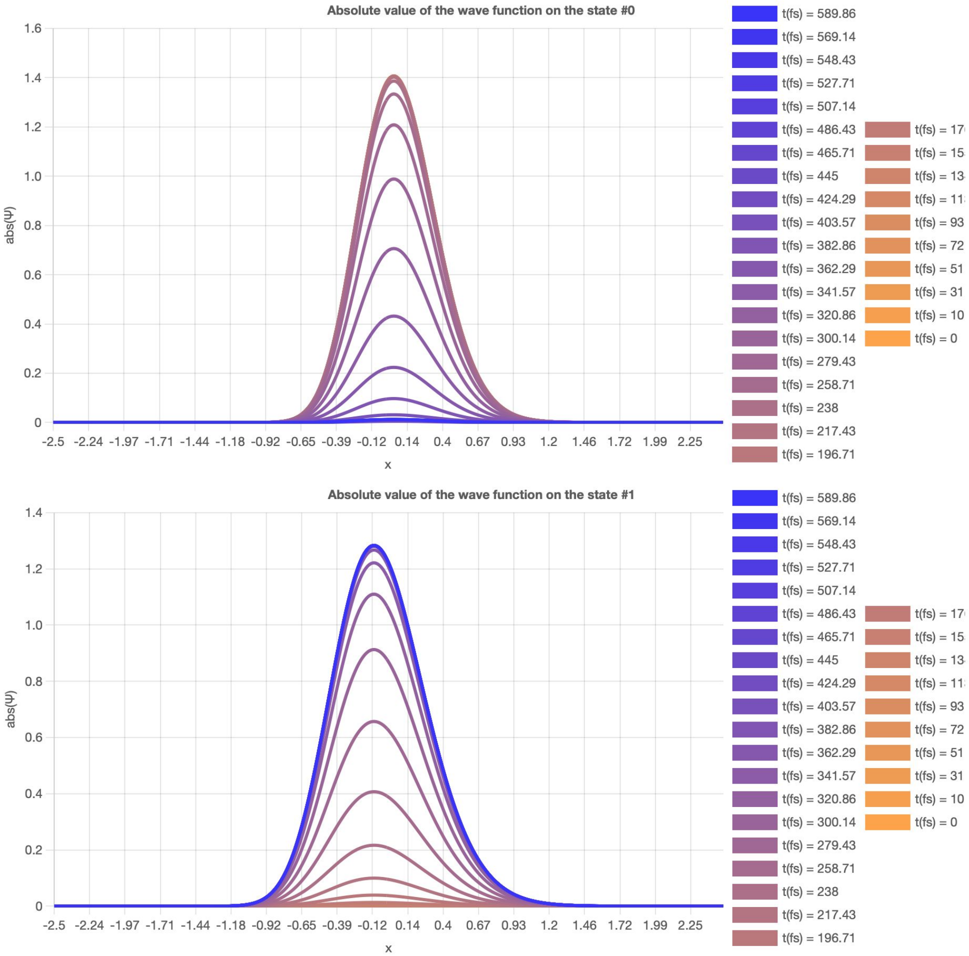
- ❖ **`task type = optimal control krotov`**
  Calculation of transition from the ground state to the excited one under the influence of a controlled external laser field envelope using an iterative Krotov algorithm

- ❖ **`task type = optimal control unit transform`**
  Calculation of a unitary transformation of a system with an Hadamard goal operator and a 2-level / Bose-Hubbard Hamiltonian in its angular momentum form, using a controlled external laser field envelope with an iterative Krotov algorithm

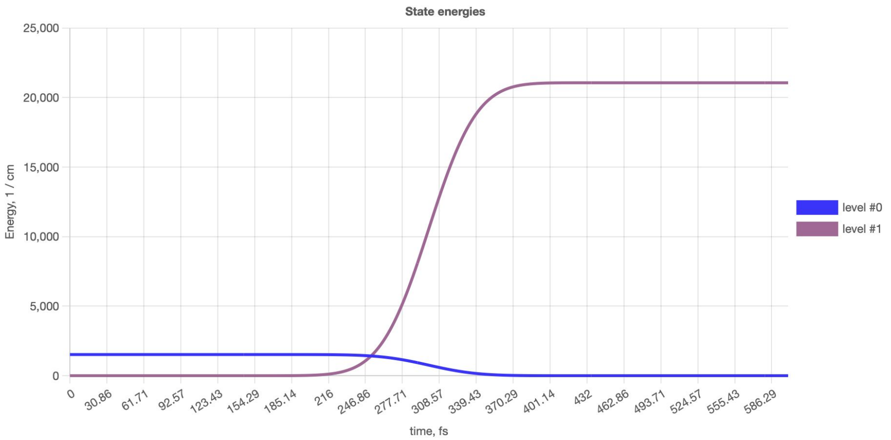
Results: Transition without control

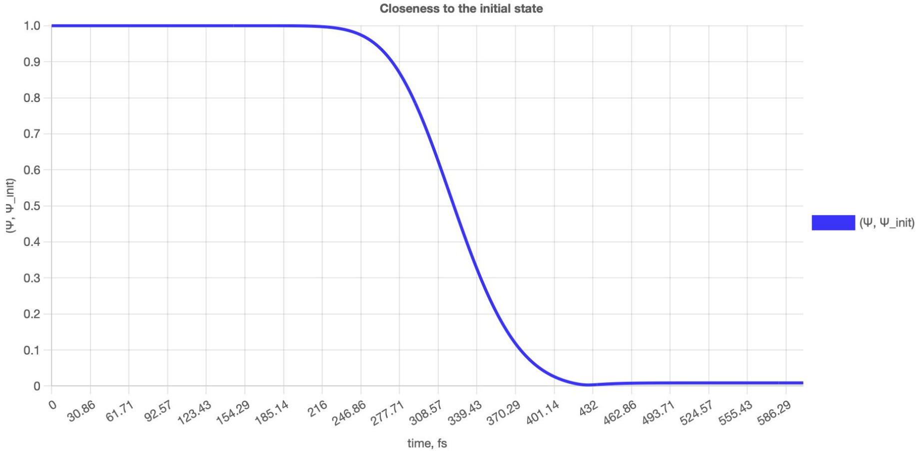
input_task.json:

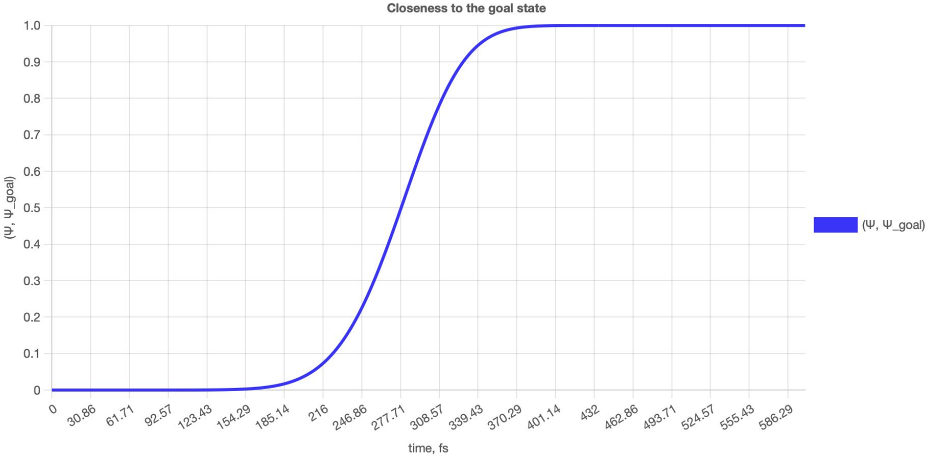
```
{
 "task_type": "trans_wo_control",
 "pot_type": "morse",
 "wf_type": "morse",
 "hamil_type": "ntriv",
 "init_guess": "gauss",
 "init_guess_hf": "exp",
 "nb": 1,
 "T": 600e-15,              // s
 "L": 5.0,                  // a_0
 "np": 1024,
 "a": 1.0,                  // 1/a_0
 "De": 20000.0,             // 1/cm
 "x0p": -0.17,              // a_0
 "a_e": 1.0,                // 1/a_0
 "De_e": 10000.0,           // 1/cm
 "Du": 20000.0,             // 1/cm
 "fitter": {
   "impulses_number": 1,
   "propagation": {
     "m": 0.5,              // Dalton
     "nch": 64,
     "nt": 420000,
     "E0": 71.54,           // 1/cm
     "t0": 300e-15,         // s
     "sigma": 50e-15,       // s
     "nu_L": 0.29297e15     // Hz
   },
   "mod_log": 500
 }
}
```



Results: Transition without control



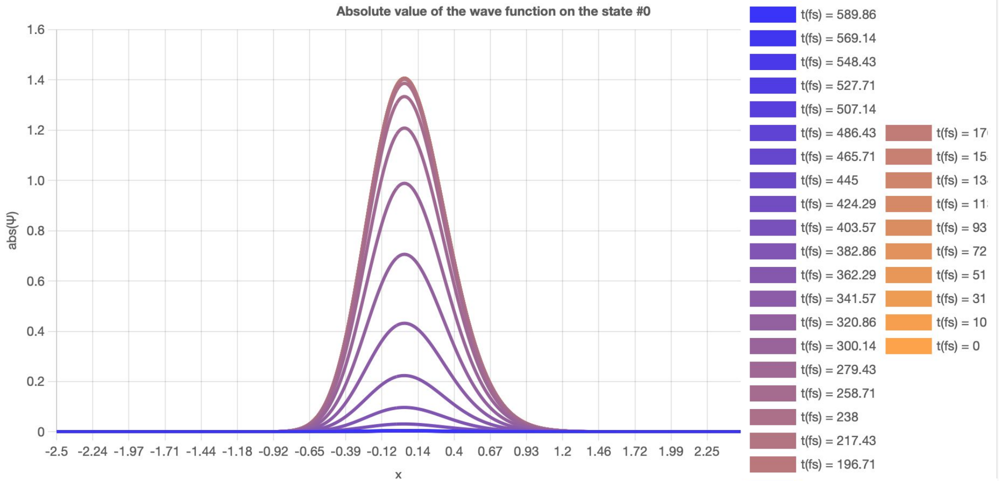


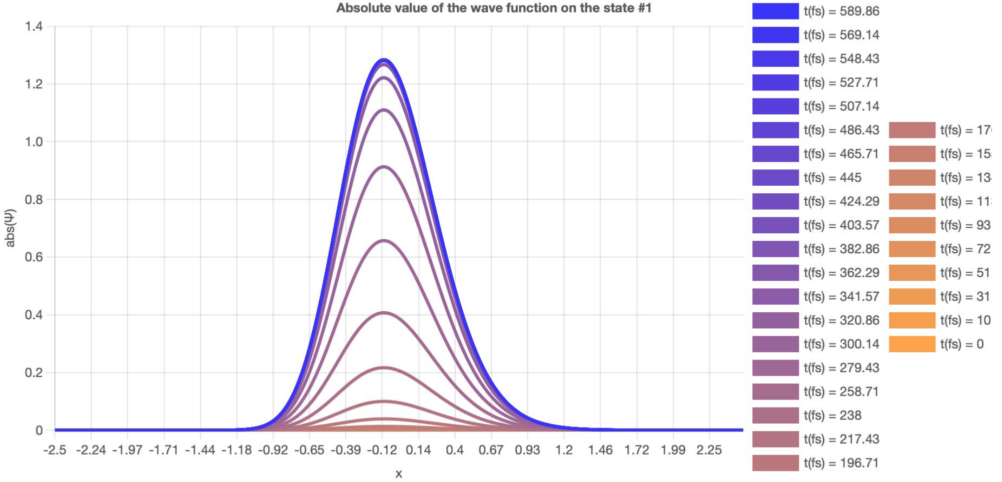


Results: local control with projection target operator

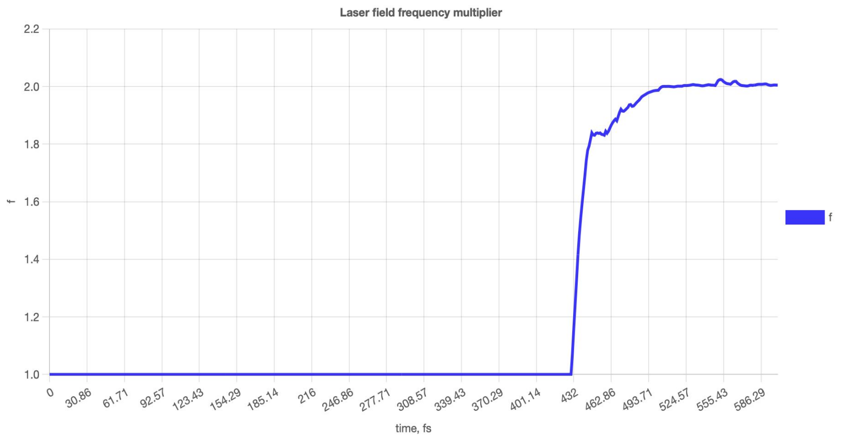
input_task.json:

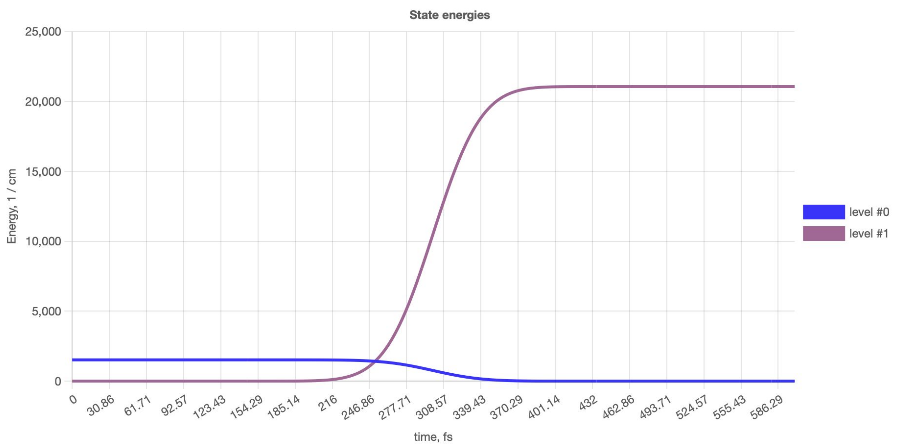
```
{
 "task_type": "local_control_projection",
 "pot_type": "morse",
 "wf_type": "morse",
 "hamil_type": "ntriv",
 "init_guess": "gauss",
 "init_guess_hf": "exp",
 "nb": 1,
 "T": 600e-15,              // s
 "L": 5.0,                  // a_0
 "np": 1024,
 "a": 1.0,                  // 1/a_0
 "De": 20000.0,             // 1/cm
 "x0p": -0.17,              // a_0
 "a_e": 1.0,                // 1/a_0
 "De_e": 10000.0,           // 1/cm
 "Du": 20000.0,             // 1/cm
 "fitter": {
   "k_E": 1e29,             // Hz*Hz
   "lamb": 8e14,            // Hz
   "pow": 0.65,
   "epsilon": 1e-15,
   "impulses_number": 1,
   "propagation": {
     "m": 0.5,              // Dalton
     "nch": 64,
     "nt": 420000,
     "E0": 71.54,           // 1/cm
     "t0": 300e-15,         // s
     "sigma": 50e-15,       // s
     "nu_L": 0.29297e15     // Hz
   },
   "mod_log": 500
 }
}
```

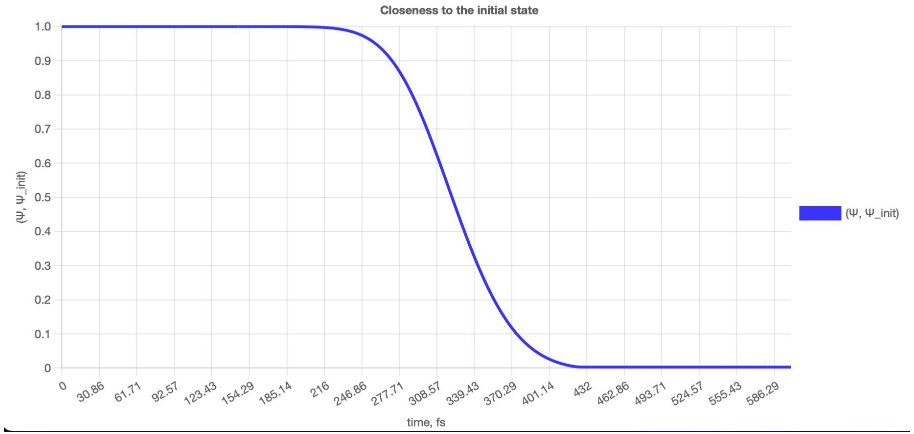


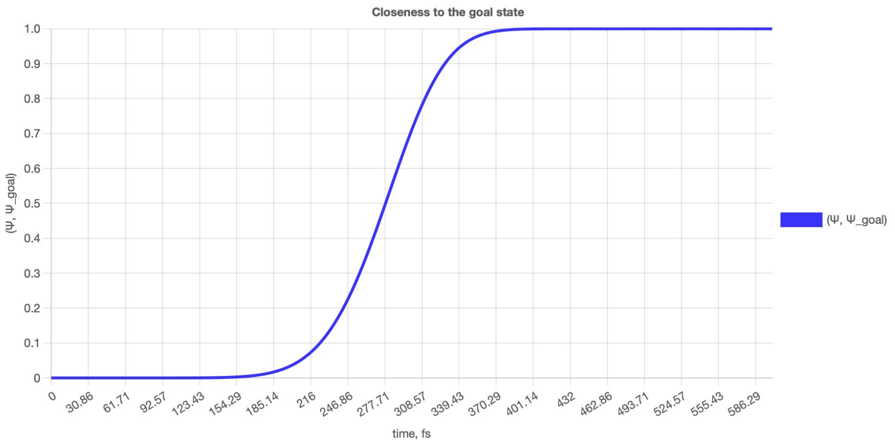


Results: local control with projection target operator





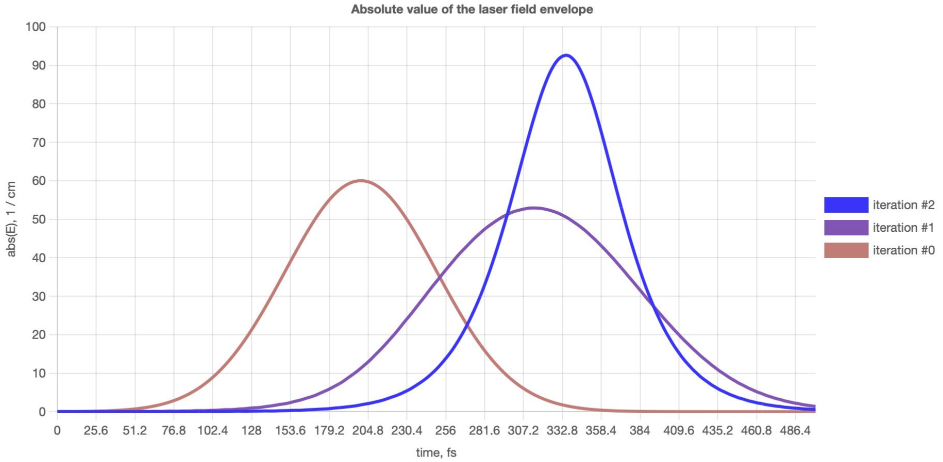


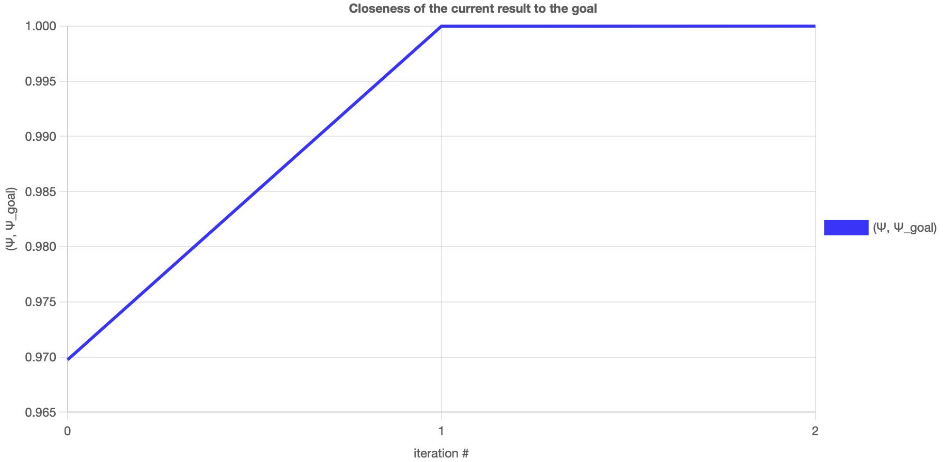


Results: Krotov optimal control in Morse potential

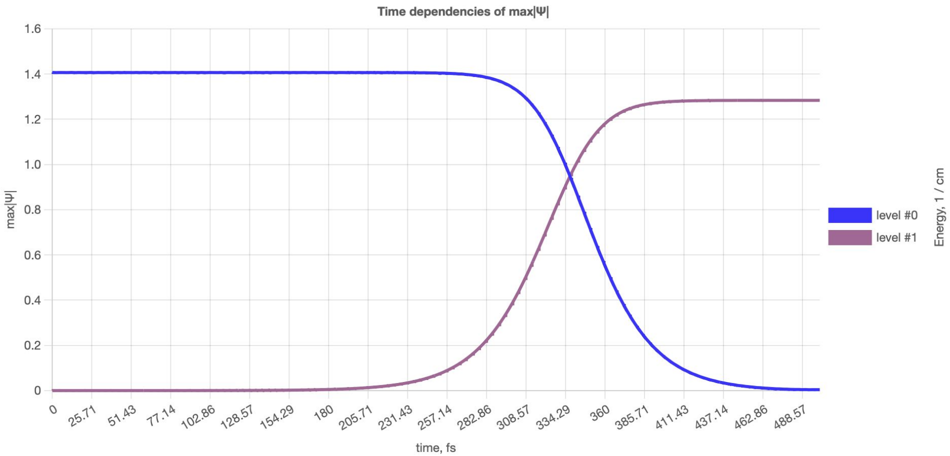
input_task.json:

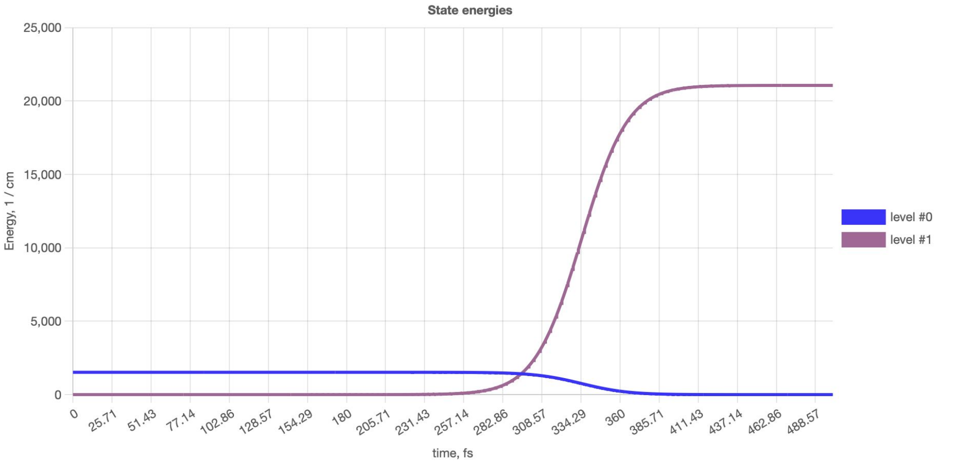
```
{
 "task_type": "optimal_control_krotov",
 "pot_type": "morse",
 "wf_type": "morse",
 "hamil_type": "ntriv",
 "init_guess": "gauss",
 "init_guess_hf": "exp",
 "nb": 1,
 "T": 500e-15,              // s
 "L": 5.0,                  // a_0
 "np": 1024,
 "a": 1.0,                  // 1/a_0
 "De": 20000.0,             // 1/cm
 "x0p": -0.17,              // a_0
 "a_e": 1.0,                // 1/a_0
 "De_e": 10000.0,           // 1/cm
 "Du": 20000.0,             // 1/cm
 "fitter": {
   "epsilon": 1e-8,
   "impulses_number": 1,
   "iter_max": 2,
   "h_lambda": 0.0066,
   "propagation": {
     "m": 0.5,              // Dalton
     "nch": 64,
     "nt": 350000,
     "E0": 60.0,            // 1/cm
     "t0": 200e-15,         // s
     "sigma": 50e-15,       // s
     "nu_L": 0.29297e15     // Hz
   },
   "mod_log": 500
 }
}
```

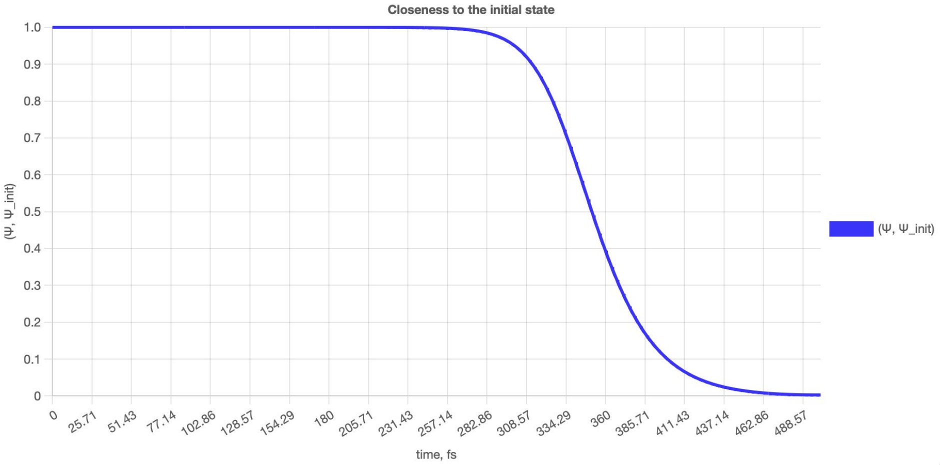


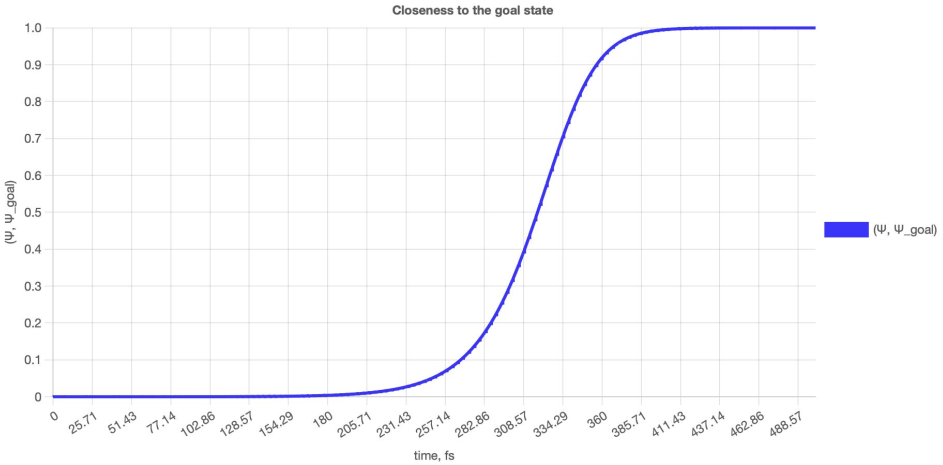


Results: Krotov optimal control in Morse potential (last iteration)





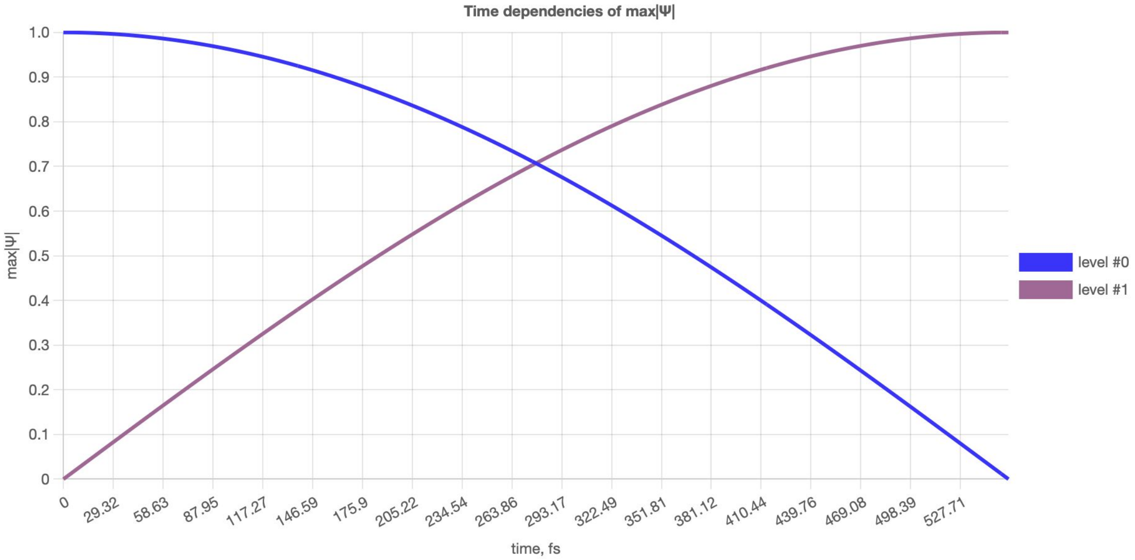




Results: Qubit excitation (from https://en.wikipedia.org/wiki/Rabi_cycle)

input_task.json:

```
{
    "task_type": "trans_wo_control",
    "hamil_type": "BH_model",
    "lf_aug_type": "x",
    "init_guess": "const",
    "init_guess_hf": "exp",
    "nb": 1,
    "nlevs": 2,
    "T": 555.9416E-15,              // s
    "U": 30.0,                      // 1/cm
    "W": 0.0,                       // 1/cm
    "delta": 15.0,                  // 1/cm
    "fitter": {
        "epsilon": 1e-5,
        "impulses_number": 1,
        "iter_max": 300,
        "iter_mid_1": 250,
        "iter_mid_2": 300,
        "q": 0.75,
        "h_lambda": 0.00005,
        "h_lambda_mode": "const",
        "F_type": "sm",
        "hf_hide": False,
        "propagation": {
            "nch": 8,
            "t0": 0.0,              // s
            "E0": 1.0,              // 1/cm
            "nt": 512,
            "nu_L": 0.00179875E15   // Hz
        },
        "mod_log": 500
    }
}
```



$$\hat{H} = \begin{pmatrix} U & -\Delta E_L^* \\ -\Delta E_L & -U \end{pmatrix} \quad U = \frac{\hbar\omega_0}{2} \quad \Delta = \frac{\hbar\omega_1}{2} \quad E_L = \exp\left(i\omega t\right)$$

$$\psi(t) = \begin{pmatrix} \exp\left(i\frac{\omega_0 t}{2}\right)\cos\left(\frac{\omega_1 t}{2}\right) \\ i\exp\left(-i\frac{\omega_0 t}{2}\right)\sin\left(\frac{\omega_1 t}{2}\right) \end{pmatrix}$$

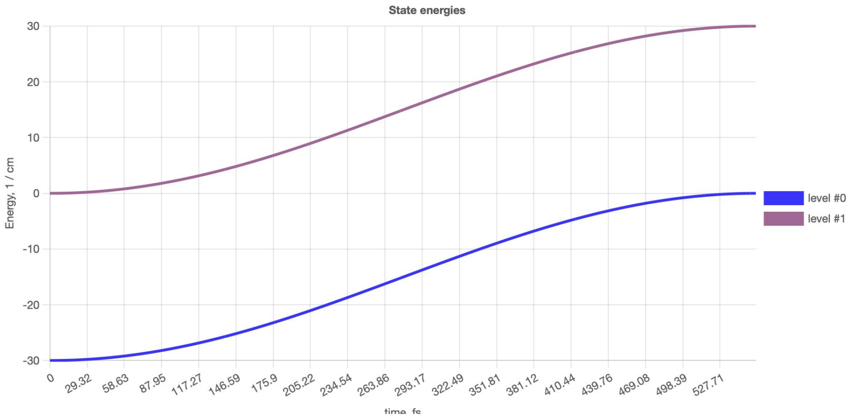
Resonance condition:

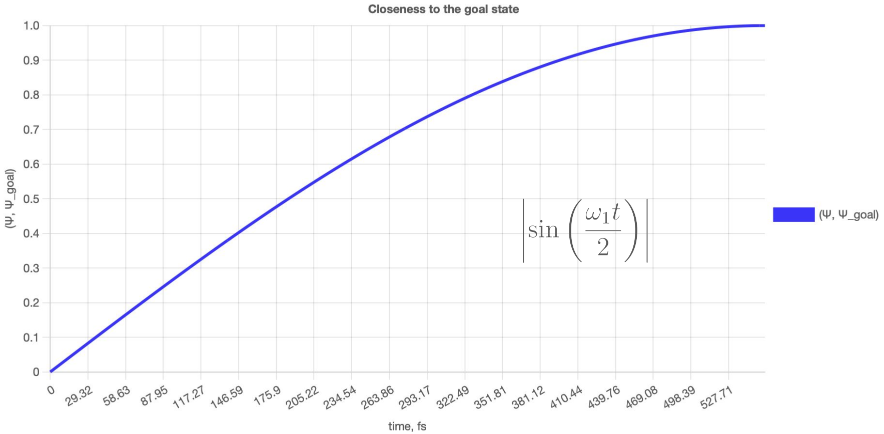
$$\omega = \omega_0 \therefore \nu_L = [2U]_{Hz}$$

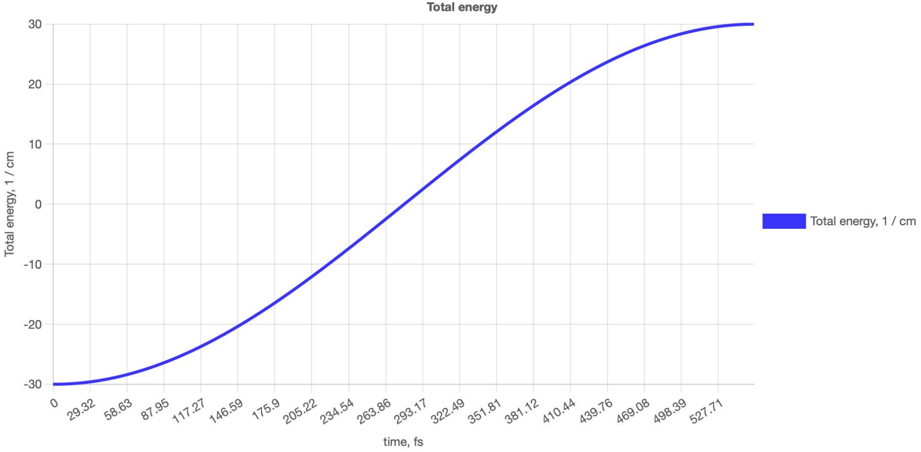
$$T = \frac{\pi}{\omega_1} \therefore T = \frac{1}{[4\Delta]_{Hz}}$$ – Pi-pulse

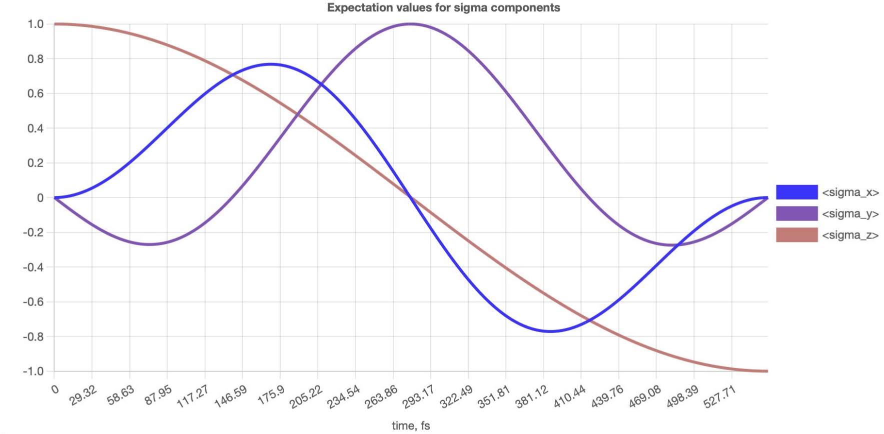
$$\omega_1 = 2\pi[2\Delta]_{Hz}$$

Results: Qubit excitation (from https://en.wikipedia.org/wiki/Rabi_cycle)





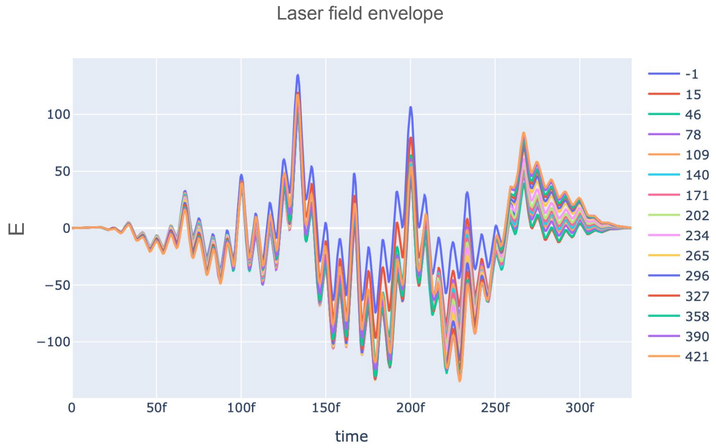


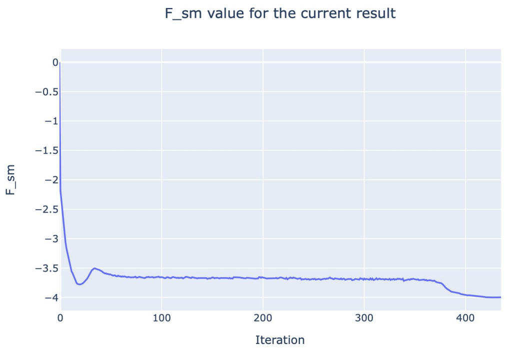


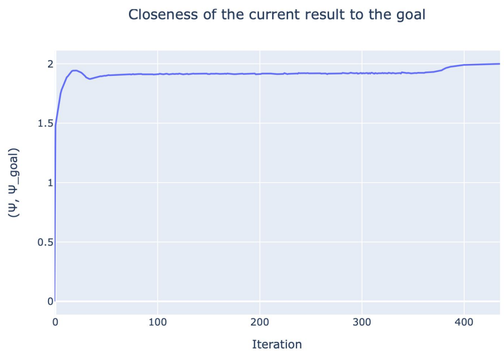
Results: unitary transitions with BH_Z Hamiltonian

input_task.json:

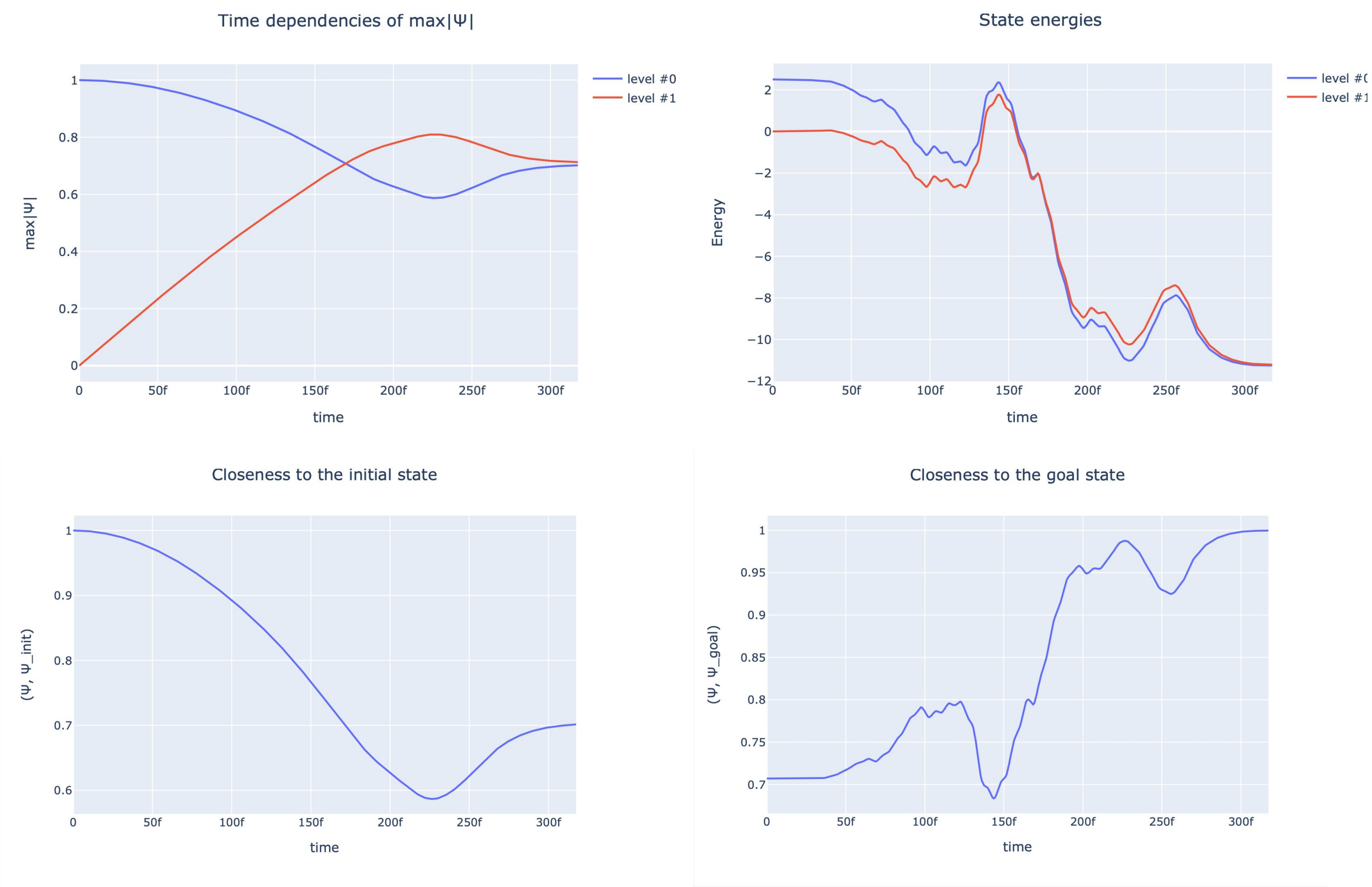
```
{
 "task_type": "optimal_control_unit_transform",
 "hamil_type": "BH_model",
 "lf_aug_type": "z",
 "init_guess": "sqrsin",
 "init_guess_hf": "cos_set",
 "nb": 2,
 "T": 3.306555E-13,                  // s
 "U": 5.0,                           // 1/cm
 "delta": 25.0,                      // 1/cm
 "fitter": {
   "epsilon": 1e-8,
   "impulses_number": 1,
   "iter_max": 430,
   "h_lambda": 0.005,
   "w_list": [
     0.88, 0.34, 0.92, 0.27, 0.39, 0.82, 0.68
   ],
   "pcos": 4.0,
   "hf_hide": false,
   "Em": 5.0,
   "propagation": {
     "nch": 8,
     "t0": 0.0,                      // s
     "E0": 40.0,                     // 1/cm
     "nt": 3500,
     "sigma": 6.671270E-13,          // s
     "nu_L": 0.299793E14             // Hz
   },
   "mod_log": 500
 }
}
```







Results: unitary transitions with BH_Z Hamiltonian (last iteration)



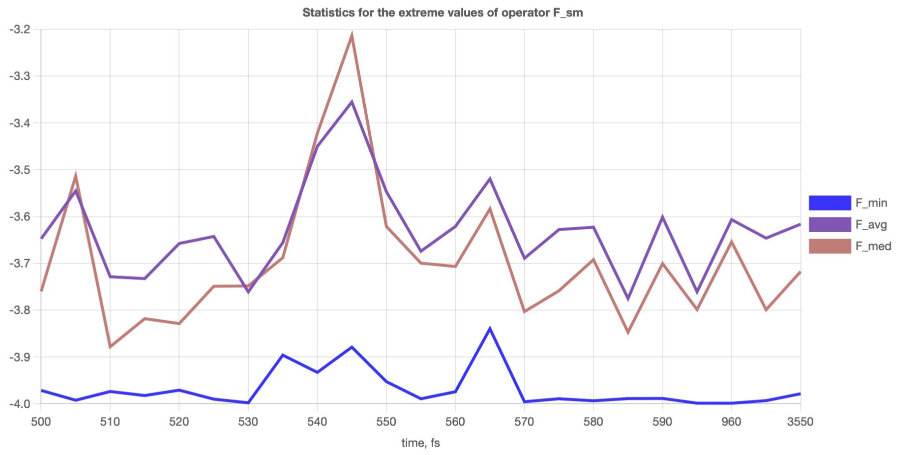
Results: batch runs – quantum window search

input_task.json:

```
{
 "run_id": { "@SUBST:LIST": [ "ut/run1", "ut/run2", "ut/run3",
"ut/run4", "ut/run5", "ut/run6", "ut/run7", "ut/run8",
"ut/run9", "ut/run10", "ut/run11", "ut/run12", "ut/run13",
"ut/run14", "ut/run15", "ut/run16", "ut/run17", "ut/run18",
"ut/run19", "ut/run20" ] },
 "task_type": "optimal_control_unit_transform",
 "hamil_type": "BH_model",
 "lf_aug_type": "x",
 "init guess": "sqrsin",
 "init guess_hf": "sin_set",
 "nb": 2,
 "T": { "@SUBST:LIST":
 [
   5.05E-13,   5.1E-13,   5.15E-13,   5.2E-13,   5.25E-13,
   5.3E-13,    5.35E-13,  5.4E-13,    5.45E-13,  5.5E-13,
   5.55E-13,   5.6E-13,   5.65E-13,   5.7E-13,   5.75E-13,
   5.8E-13,    5.85E-13,  5.9E-13,    5.95E-13,  5E-13,
   9.6E-13,    22E-13,    35.5E-13
 ] },
 "U": 30.0,                     // 1/cm
 "W": 30.0,                     // 1/cm
 "delta": 15.0,                 // 1/cm
 "sigma_auto": true,
 "nu L auto": true,
 "fitter": {
   "epsilon": 1e-3,
   "impulses_number": 1,
   "iter_max": 5000,
   "iter mid 1": 250,
   "iter mid 2": 300,
   "q": 0.75,
   "h_lambda": 0.00005,
   "h_lambda_mode":"dynamical",
   "pcos": 2.0,
   "hf hide": false,
   "propagation": {
     "nch": 8,
     "t0": 0.0,                 // s
     "E0": 40.0,                // 1/cm
     "nt": 3500
   },
   "mod_log": 500
 }
}
```

glob.txt:

```
...........
T = 570.0 fs
run107    5000      -3.995594
run93     5000      -3.96636

T = 575.0 fs
run61     5000      -3.946481
run183    4954      -3.989304

T = 580.0 fs
run45     1859      -3.993622
run71     5000      -3.967377

T = 585.0 fs
run55     2028      -3.976704
run47     4999      -3.988873
run85     5000      -3.981385

T = 590.0 fs
run160    851       -3.988568
run178    4996      -3.977092

T = 595.0 fs
run54     4837      -3.998701
run188    2469      -3.97156

T = 960.0 fs
run170    740       -3.998801
run161    2091      -3.953913

..........
```



Summary

- **Qcontrol** – a piece of software for design and optimization of pulse-level quantum control algorithms for qubit-like quantum systems
- Contains algorithms for solving Schrodinger equation for time-dependent Hamiltonian evolution of small quantum systems and modeling their behavior under a controlled laser field excitation (**quantum control problem**)
- Is a flexible **Python 3 software solution** aimed for numerical calculation of dynamics for different quantum system types, including general unitary transformations
- The software was designed to be modular, easily maintainable and expandable. In its latest version, the code supports **12+ types** of different quantum mechanical problems
- Contains a custom multi-layer data **visualization system** for the output results
- Contains a custom **DSL** for parsing complex, multi-variable configuration files
- The code is clear, well-refactored, well-documented, includes about 25 automated unit and functional tests and generally complies to good quality software product standards. It consists of about **7.5K lines of code** and contains about **60 classes**

Thank you for attention!